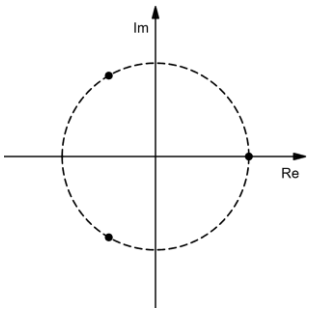
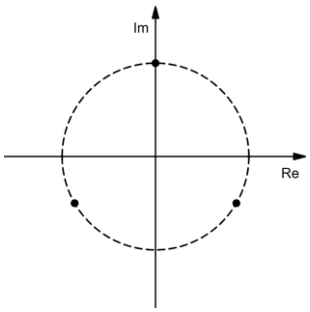
1) a)

All the roots lie on a unit circle centred at the origin:



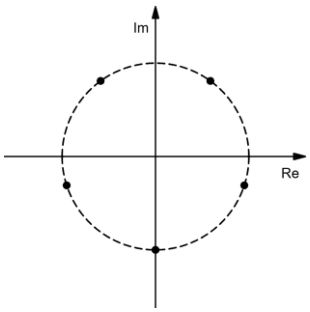
1) b)

All the roots lie on a unit circle centred at the origin:



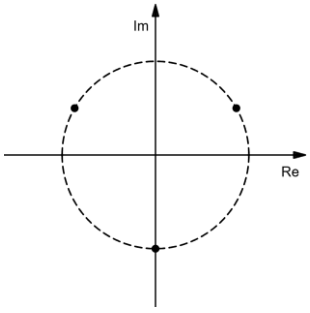
1) c)

All the roots lie on a unit circle centred at the origin:



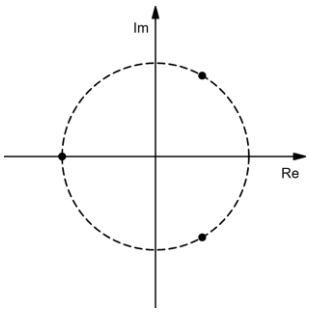
1) d)

All the roots lie on a circle of radius 2 centred at the origin:



1) e)

All the roots lie on a circle of radius 2 centred at the origin:



2) a)

$z = \mathrm{e}^{\frac{\pi}{6}\mathrm{i}}$, $z = \mathrm{e}^{\frac{5\pi}{6}\mathrm{i}}$ and $z = \mathrm{e}^{-\frac{\pi}{2}\mathrm{i}}$

$\therefore z_0 = \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$, $z_1 = -\frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}$ and $z_2 = -\mathrm{i}$

2) b)

$a = 3$, $b = 1$

3) a)

$z = 4\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$, $z = 4\mathrm{e}^{\pi\mathrm{i}}$ and $z = 4\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}}$

$\therefore z_0 = 2 + 2\sqrt{3}\mathrm{i}$, $z_1 = -4$ and $z_2 = 2 - 2\sqrt{3}\mathrm{i}$

3) b)

$a = 3$, $b = -64$

4) a)

$z_0 = \sqrt{3}\mathrm{e}^{\frac{\pi}{8}\mathrm{i}}$, $z_1 = \sqrt{3}\mathrm{e}^{\frac{5\pi}{8}\mathrm{i}}$, $z_2 = \sqrt{3}\mathrm{e}^{\frac{9\pi}{8}\mathrm{i}}$ and $z_3 = \sqrt{3}\mathrm{e}^{\frac{13\pi}{8}\mathrm{i}}$

$\therefore z = \sqrt{3}\mathrm{e}^{\frac{\pi}{8}\mathrm{i}}$, $z = \sqrt{3}\mathrm{e}^{\frac{5\pi}{8}\mathrm{i}}$, $z = \sqrt{3}\mathrm{e}^{-\frac{7\pi}{8}\mathrm{i}}$ and $z = \sqrt{3}\mathrm{e}^{-\frac{3\pi}{8}\mathrm{i}}$

4) b)

$a = 4$, $b = 9$

5) a)

$z^3 = 8$

Using modulus-argument form: $8 = 8(\cos 0 + \mathrm{i}\sin 0)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 8 \Rightarrow r = 2$

Since $3\theta = 0 + 2k\pi$ for integer k: $\theta = 0 + \frac{2k\pi}{3}$

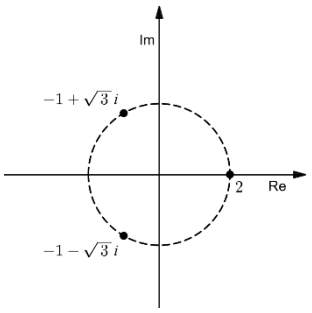
When $k = 0$: $z_0 = 2\mathrm{e}^{0\mathrm{i}} = 2$

When $k = 1$: $z_1 = 2\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} = -1 + \sqrt{3}\mathrm{i}$

When $k = 2$: $z_2 = 2\mathrm{e}^{\frac{4\pi}{3}\mathrm{i}} = 2\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}} = -1 - \sqrt{3}\mathrm{i}$

$\therefore z = 2$, $z = -1 + \sqrt{3}\mathrm{i}$ and $z = -1 - \sqrt{3}\mathrm{i}$

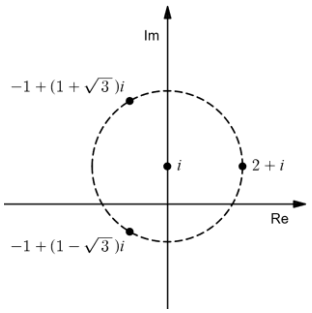
5) b)



5) c)

$z = 2 + \mathrm{i}$, $z =$ and $z = -1 + (1 - \sqrt{3})\mathrm{i}$

5) d)



6) a)

$z^4 = 16\text{i}$

Using modulus-argument form: $16\text{i} = 16\left(\cos\frac{\pi}{2} + \text{i}\sin\frac{\pi}{2}\right)$

$z^4 = r^4(\cos\theta + \text{i}\sin\theta)^4 = r^4(\cos 4\theta + \text{i}\sin 4\theta)$

$r^4 = 16 \Rightarrow r = 2$

Since $4\theta = \frac{\pi}{2} + 2k\pi$ for integer k: $\theta = \frac{\pi}{8} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = 2\text{e}^{\frac{\pi}{8}\text{i}}$

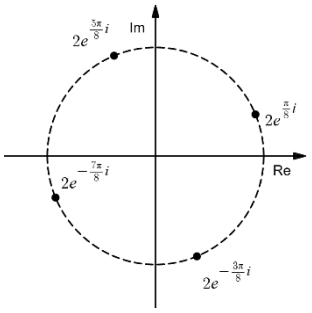
When $k = 1$: $z_1 = 2\text{e}^{\frac{5\pi}{8}\text{i}}$

When $k = 2$: $z_2 = 2\text{e}^{\frac{9\pi}{8}\text{i}} = 2\text{e}^{-\frac{7\pi}{8}\text{i}}$

When $k = 3$: $z_3 = 2\text{e}^{\frac{13\pi}{8}\text{i}} = 2\text{e}^{-\frac{3\pi}{8}\text{i}}$

$\therefore z = 2\text{e}^{\frac{\pi}{8}\text{i}}$, $z = 2\text{e}^{\frac{5\pi}{8}\text{i}}$, $z = 2\text{e}^{-\frac{7\pi}{8}\text{i}}$ and $z = 2\text{e}^{-\frac{3\pi}{8}\text{i}}$

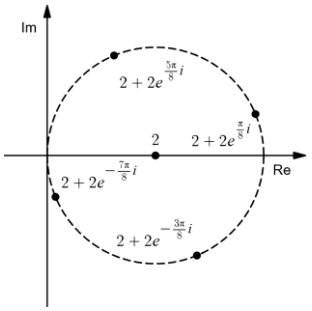
6) b)



6) c)

$z = 2 + 2\text{e}^{\frac{\pi}{8}\text{i}}$, $z = 2 + 2\text{e}^{\frac{5\pi}{8}\text{i}}$ $z = 2 + 2\text{e}^{-\frac{7\pi}{8}\text{i}}$ and $z = 2 + 2\text{e}^{-\frac{3\pi}{8}\text{i}}$

6) d)



7) a)

$z^6 = -64\text{i}$

Using modulus-argument form: $-64\text{i} = 64\left(\cos\left(-\frac{\pi}{2}\right) + \text{i}\sin\left(-\frac{\pi}{2}\right)\right)$

$z^6 = r^6(\cos\theta + \text{i}\sin\theta)^6 = r^6(\cos 6\theta + \text{i}\sin 6\theta)$

$r^6 = 64 \Rightarrow r = 2$

Since $6\theta = -\frac{\pi}{2} + 2k\pi$ for integer k: $\theta = -\frac{\pi}{12} + \frac{2k\pi}{6}$

When $k = 0$: $z_0 = 2\text{e}^{-\frac{\pi}{12}\text{i}}$

When $k = 1$: $z_1 = 2\text{e}^{\frac{\pi}{4}\text{i}}$

When $k = 2$: $z_2 = 2\text{e}^{\frac{7\pi}{12}\text{i}}$

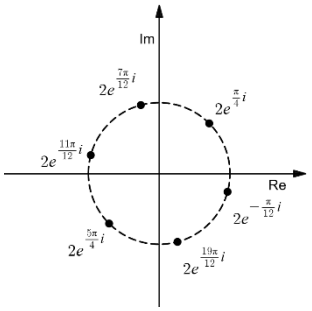
When $k = 3$: $z_3 = 2\text{e}^{\frac{11\pi}{12}\text{i}}$

When $k = 4$: $z_4 = 2\text{e}^{\frac{5\pi}{4}\text{i}} = 2\text{e}^{-\frac{\pi}{4}\text{i}}$

When $k = 5$: $z_5 = 2\text{e}^{\frac{19\pi}{12}\text{i}} = 2\text{e}^{-\frac{5\pi}{12}\text{i}}$

$\therefore z = 2\text{e}^{-\frac{\pi}{12}\text{i}}$, $z = 2\text{e}^{\frac{\pi}{4}\text{i}}$, $z = 2\text{e}^{\frac{7\pi}{12}\text{i}}$, $z = 2\text{e}^{\frac{11\pi}{12}\text{i}}$, $z = 2\text{e}^{-\frac{\pi}{4}\text{i}}$ and $z = 2\text{e}^{-\frac{5\pi}{12}\text{i}}$

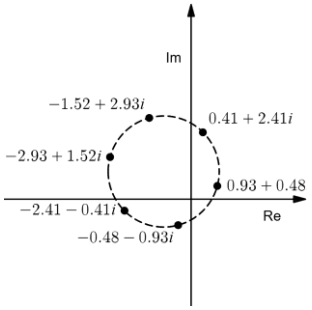
7) b)



7) c)

$z = 0.93 + 0.48\text{i}$, $z = 0.41 + 2.41\text{i}$, $z = -1.52 + 2.93\text{i}$, $z = -2.93 + 1.52\text{i}$, $z = -2.41 - 0.41\text{i}$ and $z = -0.48 - 0.93\text{i}$

7) d)



8) a)

$|1+3\text{i}| = \sqrt{1^2+3^2} = \sqrt{10}$ and $\arg(1+3\text{i}) = \arctan(3)$

To find the angle of the other two vertices: Add on $\frac{2\pi}{3}$

$\theta_1 = \arctan(3) + \frac{2\pi}{3}$ so $z_1 = \sqrt{10}\left(\cos\left(\arctan(3)+\frac{2\pi}{3}\right) + \text{i}\sin\left(\arctan(3)+\frac{2\pi}{3}\right)\right)$

$= -3.10 - 0.634\text{i}$

$\theta_2 = \arctan(3) + \frac{4\pi}{3}$ so $z_2 = \sqrt{10}\left(\cos\left(\arctan(3)+\frac{4\pi}{3}\right) + \text{i}\sin\left(\arctan(3)+\frac{4\pi}{3}\right)\right)$

$= 2.10 - 2.37\text{i}$

$\therefore (1,3)$, $(-3.10,\ -0.634)$ and $(2.10, -2.37)$

8) b)

$|3+2\text{i}| = \sqrt{9+4} = \sqrt{13}$ and $\arg(3+2\text{i}) = \arctan\left(\frac{2}{3}\right)$

To find the angle of the other two vertices: Add on $\frac{2\pi}{5}$

$\theta_1 = \arctan\left(\frac{2}{3}\right) + \frac{2\pi}{5}$ so $z_1 = \sqrt{13}\left(\cos\left(\arctan\left(\frac{2}{3}\right)+\frac{2\pi}{5}\right) + \text{i}\sin\left(\arctan\left(\frac{2}{3}\right)+\frac{2\pi}{5}\right)\right)$

$= -0.975 + 3.47\text{i}$

$\theta_2 = \arctan\left(\frac{2}{3}\right) + \frac{4\pi}{5}$ so $z_2 = \sqrt{13}\left(\cos\left(\arctan\left(\frac{2}{3}\right)+\frac{4\pi}{5}\right) + \text{i}\sin\left(\arctan\left(\frac{2}{3}\right)+\frac{4\pi}{5}\right)\right)$

$= -3.60 + 0.145\text{i}$

$\theta_3 = \arctan\left(\frac{2}{3}\right) + \frac{6\pi}{5}$ so $z_3 = \sqrt{13}\left(\cos\left(\arctan\left(\frac{2}{3}\right)+\frac{6\pi}{5}\right) + \text{i}\sin\left(\arctan\left(\frac{2}{3}\right)+\frac{6\pi}{5}\right)\right)$

$= -1.25 - 3.38\text{i}$

$\theta_4 = \arctan\left(\frac{2}{3}\right) + \frac{8\pi}{5}$ so $z_4 = \sqrt{13}\left(\cos\left(\arctan\left(\frac{2}{3}\right)+\frac{8\pi}{5}\right) + \text{i}\sin\left(\arctan\left(\frac{2}{3}\right)+\frac{8\pi}{5}\right)\right)$

$= 2.83 - 2.24\text{i}$

$\therefore (3,2)$, $(-0.975, 3.47)$,$(-3.60, 0.145)$, $(-1.25, -3.38)$ and $(2.83, -2.24)$

8) c)

$|2+2\text{i}| = \sqrt{4+4} = 2\sqrt{2}$ and $\arg(2+2\text{i}) = \arctan(1)$

To find the angle of the other two vertices: Add on $\frac{2\pi}{6}$

$\theta_1 = \arctan(1) + \frac{2\pi}{6}$ so $z_1 = 2\sqrt{2}\left(\cos\left(\arctan(1)+\frac{2\pi}{6}\right) + \text{i}\sin\left(\arctan(1)+\frac{2\pi}{6}\right)\right)$

$= -0.732 + 2.73\text{i}$

$\theta_2 = \arctan(1) + \frac{4\pi}{6}$ so $z_2 = 2\sqrt{2}\left(\cos\left(\arctan(1)+\frac{4\pi}{6}\right) + \text{i}\sin\left(\arctan(1)+\frac{4\pi}{6}\right)\right)$

$= -2.73 + 0.732\text{i}$

$\theta_3 = \arctan(1) + \frac{6\pi}{6}$ so $z_3 = 2\sqrt{2}\left(\cos\left(\arctan(1)+\frac{6\pi}{6}\right) + \text{i}\sin\left(\arctan(1)+\frac{6\pi}{6}\right)\right)$

$= -2 - 2\text{i}$

$\theta_4 = \arctan(1) + \frac{8\pi}{6}$ so $z_4 = 2\sqrt{2}\left(\cos\left(\arctan(1)+\frac{8\pi}{6}\right) + \text{i}\sin\left(\arctan(1)+\frac{8\pi}{6}\right)\right)$

$= 0.732 - 2.73\text{i}$

$\theta_5 = \arctan(1) + \frac{10\pi}{6}$ so $z_5 = 2\sqrt{2}\left(\cos\left(\arctan(1) + \frac{10\pi}{6}\right) + \text{i}\sin\left(\arctan(1) + \frac{10\pi}{6}\right)\right)$

$= 2.73 - 0.732\text{i}$

$\therefore (2, 2)$, $(-0.732, 2.73)$,$(-2.73, 0.732)$, $(-2, -2)$, $(0.732,\ -2.73)$ and $(2.73,\ -0.732)$

9) a)

$|2 + 2\text{i}| = \sqrt{4 + 4} = 2\sqrt{2}$

Angle between the vertices: $\frac{2\pi}{3}$

Area is

$3 \times \frac{1}{2} \times 2\sqrt{2} \times 2\sqrt{2} \times \sin\left(\frac{2\pi}{3}\right) = 6\sqrt{3}$ units^2

9) b)

$|5 + \text{i}| = \sqrt{25 + 1} = \sqrt{26}$

Angle between the vertices: $\frac{2\pi}{4}$

Area is

$4 \times \frac{1}{2} \times \sqrt{26} \times \sqrt{26} \times \sin\left(\frac{2\pi}{4}\right) = 52$ units^2

9) c)

$|3 + 4\text{i}| = \sqrt{9 + 16} = 5$

Angle between the vertices: $\frac{2\pi}{6}$

Area is

$6 \times \frac{1}{2} \times 5 \times 5 \times \sin\left(\frac{2\pi}{6}\right) = \frac{75}{2}\sqrt{3}$ units^2

10)

Translate $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$ so vertex at $(2, 2)$ becomes vertex at $(2, -1)$

$|2 - \mathrm{i}| = \sqrt{4 + 1} = \sqrt{5}$ and $\arg(2 - \mathrm{i}) = -\arctan\left(\frac{1}{2}\right)$

To find the angle of the other two vertices: Add on $\frac{2\pi}{6}$

$\theta_1 = -\arctan\left(\frac{1}{2}\right) + \frac{2\pi}{6}$ so $z_1 = \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{2\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{2\pi}{6}\right)\right)$

$= 1.87 + 1.23\mathrm{i}$

$\theta_2 = -\arctan\left(\frac{1}{2}\right) + \frac{4\pi}{6}$ so

$z_2 = \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{4\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{4\pi}{6}\right)\right) = -0.13 + 2.23\mathrm{i}$

$\theta_3 = -\arctan\left(\frac{1}{2}\right) + \frac{6\pi}{6}$ so $z_3 = \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{6\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{6\pi}{6}\right)\right)$

$= -2 + \mathrm{i}$

$\theta_4 = -\arctan\left(\frac{1}{2}\right) + \frac{8\pi}{6}$ so $z_4 = \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{8\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{8\pi}{6}\right)\right)$

$= -1.87 - 1.23\mathrm{i}$

$\theta_5 = -\arctan\left(\frac{1}{2}\right) + \frac{10\pi}{6}$ so

$z_5 = \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{10\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{10\pi}{6}\right)\right) = 0.13 - 2.23\mathrm{i}$

Translate points back:

$\therefore (2, 2)$, $(1.87, 4.23)$,$(-0.13, 5.23)$, $(-2, 4)$, $(-1.87, 1.77)$ and $(0.13, 0.77)$

11)

Translate $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$ so vertex at $(2, -1)$ becomes vertex at $(1, -5)$

$|1 - 5\mathrm{i}| = \sqrt{1 + 25} = \sqrt{26}$ and $\arg(1 - 5\mathrm{i}) = -\arctan 5$

To find the angle of the other two vertices: Add on $\frac{2\pi}{3}$

$\theta_1 = -\arctan 5 + \frac{2\pi}{3}$ so $z_1 = \sqrt{26}\left(\cos\left(-\arctan 5 + \frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\arctan 5 + \frac{2\pi}{3}\right)\right)$

$= 3.83 + 3.37\mathrm{i}$

$\theta_2 = -\arctan 5 + \frac{4\pi}{3}$ so $z_2 = \sqrt{26}\left(\cos\left(-\arctan 5 + \frac{4\pi}{3}\right) + \mathrm{i}\sin\left(-\arctan 5 + \frac{4\pi}{3}\right)\right)$

$= -4.83 + 1.63\mathrm{i}$

Translate points back:

$\therefore (2, -1)$, $(4.83, 7.37)$ and $(-3.83, 5.63)$

12)

$3\sqrt{3} = 3 \times \frac{1}{2}x^2 \sin\left(\frac{2\pi}{3}\right)$ so $x = 2$

Therefore $(2, 0)$ is a vertex of the triangle.

$\theta_1 = \frac{2\pi}{3}$ so $z_1 = 2\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) = -1 + \sqrt{3}\mathrm{i}$

$\theta_2 = \frac{4\pi}{3}$ so $z_2 = 2\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right) = -1 - \sqrt{3}\mathrm{i}$

$\therefore (2, 0)$, $\left(-1, \sqrt{3}\right)$ and $\left(-1, -\sqrt{3}\right)$

13)

$12\sqrt{3} = 3 \times \frac{1}{2}x^2 \sin\left(\frac{2\pi}{3}\right)$ so $x = 4$

Therefore $(1, 5)$ is a vertex of the triangle.

Translate $\begin{pmatrix}-1\\-1\end{pmatrix}$ so vertex at $(1, 5)$ becomes vertex at $(0, 4)$

$|4\mathrm{i}| = \sqrt{0 + 16} = 4$ and $\arg(4\mathrm{i}) = \frac{\pi}{2}$

To find the angle of the other two vertices: Add on $\frac{2\pi}{3}$

$\theta_1 = \frac{\pi}{2} + \frac{2\pi}{3}$ so $z_1 = 4\left(\cos\frac{7\pi}{6} + \mathrm{i}\sin\frac{7\pi}{6}\right) = -2\sqrt{3} - 2\mathrm{i}$

$\theta_2 = \frac{\pi}{2} + \frac{4\pi}{3}$ so $z_2 = 4\left(\cos\frac{11\pi}{6} + \mathrm{i}\sin\frac{11\pi}{6}\right) = 2\sqrt{3} - 2\mathrm{i}$

Translate points back:

$\therefore (1, 5)$, $\left(1 - 2\sqrt{3},\ -1\right)$ and $\left(1 + 2\sqrt{3},\ -1\right)$

14) a)

$|3+4\mathrm{i}| = \sqrt{9+16} = 5$ and $\arg(3+4\mathrm{i}) = \arctan\frac{4}{3}$

To find the angle of the other two vertices: Add on $\frac{2\pi}{3}$

$\theta_1 = \arctan\frac{4}{3} + \frac{2\pi}{3}$ so $z_1 = 5\left(\cos\left(\arctan\frac{4}{3} + \frac{2\pi}{3}\right) + \mathrm{i}\sin\left(\arctan\frac{4}{3} + \frac{2\pi}{3}\right)\right)$

$= -4.96 + 0.60\mathrm{i}$

$\theta_2 = \arctan\frac{4}{3} + \frac{4\pi}{3}$ so $z_2 = 5\left(\cos\left(\arctan\frac{4}{3} + \frac{4\pi}{3}\right) + \mathrm{i}\sin\left(\arctan\frac{4}{3} + \frac{4\pi}{3}\right)\right)$

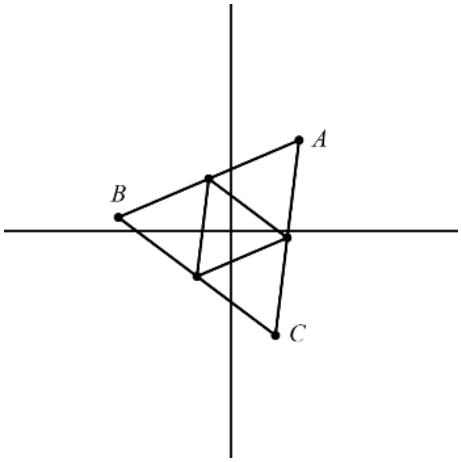
$= 1.96 - 4.60\mathrm{i}$

$\therefore B(-4.96, 0.60)$ and $C(1.96,\ -4.60)$

14) b)

$3 \times \frac{1}{2} \times 5^2 \times \sin\left(\frac{2\pi}{3}\right) = \frac{75}{4}\sqrt{3}$ units^2

14) c)



$\frac{1}{4} \times 3 \times \frac{1}{2} \times 5^2 \times \sin\left(\frac{2\pi}{3}\right) = \frac{75}{16}\sqrt{3}$ units^2

15)

$3 \times \frac{1}{2} \times x^2 \times \sin\left(\frac{2\pi}{3}\right) = 27\sqrt{3} \Rightarrow x = 6$

So the roots are $z_1 = 6\mathrm{e}^{\frac{\pi}{3}\mathrm{i}} = 3 + 3\sqrt{3}\mathrm{i}$, $z_2 = 6\mathrm{e}^{\pi\mathrm{i}} = -6$ and $z_3 = 6\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}} = 3 + 3\sqrt{3}\mathrm{i}$

So $\mathrm{f}(z) = (z+6)\left(z - \left(3 + 3\sqrt{3}\mathrm{i}\right)\right)\left(z - \left(3 - 3\sqrt{3}\mathrm{i}\right)\right) = (z+6)(z^2 - 6z + 36)$

$= z^3 + 216$

$\therefore a = 216$

16)

$3 \times \frac{1}{2} \times x^2 \times \sin\left(\frac{2\pi}{3}\right) = 3\sqrt{3} \Rightarrow x = 2$

Since $z_1 = 3 + \mathrm{i}$

Translate $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$ so vertex at $(3, 1)$ becomes vertex at $(2, 0)$

$\theta_1 = 0 + \frac{2\pi}{3}$ so $z_1 = 2\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) = -1 + \sqrt{3}\mathrm{i}$

$\theta_2 = 0 + \frac{4\pi}{3}$ so $z_2 = 2\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right) = -1 - \sqrt{3}\mathrm{i}$

Translate points back:

$\therefore (3, 1)$, $\left(0, 1 + \sqrt{3}\right)$ and $\left(0, 1 - \sqrt{3}\right)$

$z_1 = 3 + \mathrm{i}$, $z_2 = \left(1 + \sqrt{3}\right)\mathrm{i}$ and $z_3 = \left(1 - \sqrt{3}\right)\mathrm{i}$

So $\mathrm{f}(z) = \left(z - (3 + \mathrm{i})\right)\left(z - \left(\left(1 + \sqrt{3}\right)\mathrm{i}\right)\right)\left(z - \left(\left(1 - \sqrt{3}\right)\mathrm{i}\right)\right)$

$= \left(z - (3 + \mathrm{i})\right)(z^2 - 2\mathrm{i}z + 2)$

$= z^3 - (3 + 3\mathrm{i})z^2 + 6\mathrm{i}z - (6 + 2\mathrm{i})$

$\therefore a = -3 - 3\mathrm{i}$, $b = 6\mathrm{i}$ and $c = -6 - 2\mathrm{i}$

17)

$3 \times \frac{1}{2} \times x^2 \times \sin\left(\frac{2\pi}{3}\right) = \frac{27}{4}\sqrt{3} \Rightarrow x = 3$

Since $z_1 = 6 + 0\mathrm{i}$

Translate $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ so vertex at $(6, 0)$ becomes vertex at $(3, 0)$

$\theta_1 = 0 + \frac{2\pi}{3}$ so $z_1 = 3\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) = -\frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}$

$\theta_2 = 0 + \frac{4\pi}{3}$ so $z_2 = 3\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right) = -\frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}$

Translate points back:

$z_0 = 6 + 0\mathrm{i}$, $z_1 = \frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}$ and $z_2 = \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}$

So $\mathrm{f}(z) = \left(z - \left(\frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z - \left(\frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}\right)\right)(z - 6)$

$= (z^2 - 3z + 9)(z - 6)$

$= z^3 - 9z^2 + 27z - 54$

$\therefore a = -9$, $b = 27$ and $c = -54$

18) a)

Solve $w^3 = 27\text{i}$ where $w = z - 1$

Using modulus-argument form: $27\text{i} = 27\left(\cos\frac{\pi}{2} + \text{i}\sin\frac{\pi}{2}\right)$

$z^3 = r^3(\cos\theta + \text{i}\sin\theta)^3 = r^3(\cos 3\theta + \text{i}\sin 3\theta)$

$r^3 = 27 \Rightarrow r = 3$

Since $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k: $\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 3\left(\cos\frac{\pi}{6} + \text{i}\sin\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{2} + \frac{3}{2}\text{i}$

When $k = 1$: $z_1 = 3\left(\cos\frac{5\pi}{6} + \text{i}\sin\frac{5\pi}{6}\right) = -\frac{3\sqrt{3}}{2} + \frac{3}{2}\text{i}$

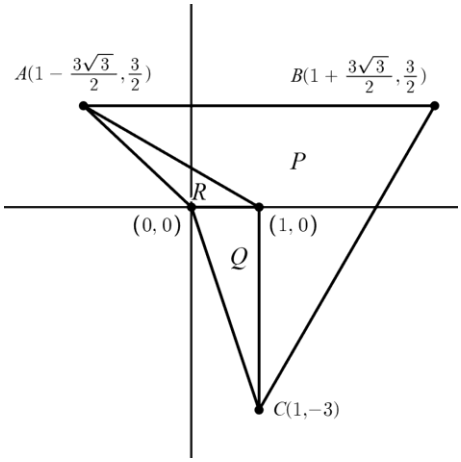
When $k = 2$: $z_2 = 3\left(\cos\left(-\frac{\pi}{2}\right) + \text{i}\sin\left(-\frac{\pi}{2}\right)\right) = -3\text{i}$

So $w = \frac{3\sqrt{3}}{2} + \frac{3}{2}\text{i}$, $w = \frac{3\sqrt{3}}{2} + \frac{3}{2}\text{i}$ and $w = -3\text{i}$ and so $z = w + 1$

$\therefore w = 1 + \frac{3\sqrt{3}}{2} + \frac{3}{2}\text{i}$, $w = 1 - \frac{3\sqrt{3}}{2} + \frac{3}{2}\text{i}$ and $w = 1 - 3\text{i}$

18) b)

Centre is at $(1, 0)$ with vertices $\left(1 + \frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$, $\left(1 - \frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$ and $(1, -3)$



Area P: $2 \times \frac{1}{2} \times 3^2 \times \sin\frac{2\pi}{3} = \frac{9\sqrt{3}}{2}$

Area Q: $\frac{1}{2} \times 1 \times 3 = \frac{3}{2}$

Area R: $\frac{1}{2} \times 1 \times \frac{3}{2} = \frac{3}{4}$

Total area: $\frac{9\sqrt{3}}{2} + \frac{9}{4}$ as req.

19) a)

$z^3 = -5 + \sqrt{75}\text{i}$

Using modulus-argument form: $-5 + \sqrt{75}\text{i} = 10\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right)$

$z^3 = r^3(\cos\theta + \text{i}\sin\theta)^3 = r^3(\cos 3\theta + \text{i}\sin 3\theta)$

$r^3 = 10 \Rightarrow r = 10^{\frac{1}{3}}$

Since $3\theta = \frac{2\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{2\pi}{9} + \frac{2k\pi}{3}$

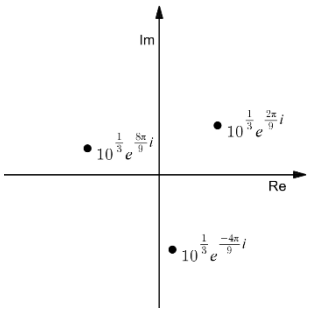
When $k = 0$: $z_0 = 10^{\frac{1}{3}}\text{e}^{\frac{2\pi}{9}\text{i}}$

When $k = 1$: $z_1 = 10^{\frac{1}{3}}\text{e}^{\frac{8\pi}{9}\text{i}}$

When $k = 2$: $z_2 = 10^{\frac{1}{3}}\text{e}^{\frac{14\pi}{9}\text{i}} = 10^{\frac{1}{3}}\text{e}^{-\frac{4\pi}{9}\text{i}}$

$\therefore z = 10^{\frac{1}{3}}\text{e}^{\frac{2\pi}{9}\text{i}}$, $z = 10^{\frac{1}{3}}\text{e}^{\frac{8\pi}{9}\text{i}}$ and $z = 10^{\frac{1}{3}}\text{e}^{-\frac{4\pi}{9}\text{i}}$

19) b)



19) c)

$\frac{2\pi}{9}$, $\frac{8\pi}{9}$ and $-\frac{4\pi}{9}$

20)

$\frac{\sqrt{3}}{2}(1 + \text{i}) = \sqrt{\frac{3}{2}}\left(\cos\frac{\pi}{4} + \text{i}\sin\frac{\pi}{4}\right)$ or $\frac{\sqrt{6}}{2}\left(\cos\frac{\pi}{4} + \text{i}\sin\frac{\pi}{4}\right)$

$\therefore$ the least value $\arg(z) = \frac{\frac{\pi}{4}}{5} = \frac{\pi}{20}$

21)

$-\frac{1}{2} + \frac{\sqrt{3}}{2}\text{i} = \cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}$

$\therefore$ the least value $\arg(z) = \frac{\frac{2\pi}{3}}{6} = \frac{\pi}{9}$

22) a)

$\mathrm{i} = \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}$

Since $6\theta = \frac{\pi}{2} + 2k\pi$ for integer k: $\theta = \frac{\pi}{12} + \frac{2k\pi}{6}$

When $k = 0$: z_0 has argument $\frac{\pi}{12}$

When $k = 1$: z_1 has argument $\frac{5\pi}{12}$

Mid-angle of z_0 and z_1 is $\frac{1}{2}\left(\frac{\pi}{12} + \frac{5\pi}{12}\right) = \frac{\pi}{4}$

Since $w = \frac{1}{2}(z_1 + z_2)$, $|z_1| = |z_2|$ and their arguments are symmetric about their midpoint, the sum lies in the direction of the angle bisector, so that

$\arg\omega = \frac{\pi}{4}$

22) b)

For ω^n to be real $\frac{n\pi}{4} = k\pi$ for some integer $k \Rightarrow n = 4k$ so let $k = 1$

$\therefore n = 4$

23) a)

$\sqrt{3} + \mathrm{i} = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

Since $5\theta = \frac{\pi}{6} + 2k\pi$ for integer k: $\theta = \frac{\pi}{30} + \frac{2k\pi}{5}$

When $k = 0$: z_0 has argument $\frac{\pi}{30}$

When $k = 1$: z_1 has argument $\frac{13\pi}{30}$

Mid-angle of z_0 and z_1 is $\frac{1}{2}\left(\frac{\pi}{30} + \frac{13\pi}{30}\right) = \frac{7\pi}{30}$

Since $w = \frac{1}{2}(z_1 + z_2)$, $|z_1| = |z_2|$ and their arguments are symmetric about their midpoint, the sum lies in the direction of the angle bisector, so that

$\arg\omega = \frac{7\pi}{30}$

23) b)

For ω^n to be real $\frac{7n\pi}{30} = k\pi$ for some integer $k \Rightarrow n = \frac{30k}{7}$ so let $k = 7$

$\therefore n = 30$

24) a)

$z^3 = \sqrt{6} + \sqrt{2}\mathrm{i}$

Using modulus-argument form: $\sqrt{6} + \sqrt{2}\mathrm{i} = 2\sqrt{2}\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = 2\sqrt{2} \Rightarrow r = \left(2\sqrt{2}\right)^{\frac{1}{3}} = \sqrt{2}$

Since $3\theta = \frac{\pi}{6} + 2k\pi$ for integer k: $\theta = \frac{\pi}{18} + \frac{2k\pi}{3}$

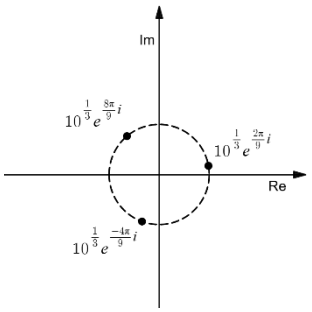
When $k = 0$: $z_0 = \sqrt{2}\mathrm{e}^{\frac{\pi}{18}\mathrm{i}}$

When $k = 1$: $z_1 = \sqrt{2}\mathrm{e}^{\frac{13\pi}{18}\mathrm{i}}$

When $k = 2$: $z_2 = \sqrt{2}\mathrm{e}^{\frac{25\pi}{18}\mathrm{i}} = \sqrt{2}\mathrm{e}^{-\frac{11\pi}{18}\mathrm{i}}$

$\therefore z = \sqrt{2}\mathrm{e}^{\frac{\pi}{18}\mathrm{i}}$, $z = \sqrt{2}\mathrm{e}^{\frac{13\pi}{18}\mathrm{i}}$ and $z = \sqrt{2}\mathrm{e}^{-\frac{11\pi}{18}\mathrm{i}}$

24) b)



24) c)

$\det \mathbf{M} = 24 - 4 = 20$

Area of triangle formed by roots: $3 \times \frac{1}{2} \times \left(\sqrt{2}\right)^2 \sin\frac{2\pi}{3} = \frac{3}{2}\sqrt{3}$

So area after transformation is $20 \times \frac{3}{2}\sqrt{3} = 30\sqrt{3}$ units^2

25) a)

$z^3 = 12\sqrt{3} + 36\text{i}$

Using modulus-argument form: $12\sqrt{3} + 36\text{i} = 24\sqrt{3}\left(\cos\frac{\pi}{3} + \text{i}\sin\frac{\pi}{3}\right)$

$z^3 = r^3(\cos\theta + \text{i}\sin\theta)^3 = r^3(\cos 3\theta + \text{i}\sin 3\theta)$

$r^3 = 24\sqrt{3} \Rightarrow r = \left(24\sqrt{3}\right)^{\frac{1}{3}} = 2\sqrt{3}$

Since $3\theta = \frac{\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{\pi}{9} + \frac{2k\pi}{3}$

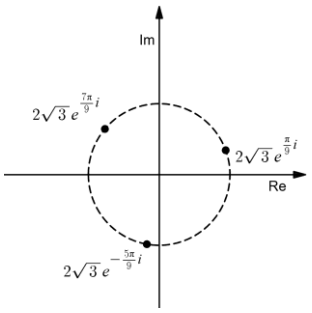
When $k = 0$: $z_0 = 2\sqrt{3}\text{e}^{\frac{\pi}{9}\text{i}}$

When $k = 1$: $z_1 = 2\sqrt{3}\text{e}^{\frac{7\pi}{9}\text{i}}$

When $k = 2$: $z_2 = 2\sqrt{3}\text{e}^{\frac{13\pi}{9}\text{i}} = 2\sqrt{3}\text{e}^{-\frac{5\pi}{9}\text{i}}$

$\therefore z = 2\sqrt{3}\text{e}^{\frac{\pi}{9}\text{i}}$, $z = 2\sqrt{3}\text{e}^{\frac{7\pi}{9}\text{i}}$ and $z = 2\sqrt{3}\text{e}^{-\frac{5\pi}{9}\text{i}}$

25) b)



25) c)

$\det\mathbf{M} = -12 + 4 = -8$

Area of triangle formed by roots: $3 \times \frac{1}{2} \times \left(2\sqrt{3}\right)^2 \sin\frac{2\pi}{3} = 9\sqrt{3}$

So area after transformation is $8 \times 9\sqrt{3} = 72\sqrt{3}$ units^2

26) a)

$z^6 = 27$

Using modulus-argument form: $27 = 27(\cos 0 + \text{i}\sin 0)$

$z^6 = r^6(\cos\theta + \text{i}\sin\theta)^6 = r^6(\cos 6\theta + \text{i}\sin 6\theta)$

$r^6 = 27 \Rightarrow r = (27)^{\frac{1}{6}} = \sqrt{3}$

Since $6\theta = 0 + 2k\pi$ for integer k: $\theta = 0 + \frac{2k\pi}{6}$

When $k = 0$: $z_0 = \sqrt{3}(\cos 0 + \text{i}\sin 0) = \sqrt{3}$

When $k = 1$: $z_1 = \sqrt{3}\left(\cos\frac{\pi}{3} + \text{i}\sin\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} + \frac{3}{2}\text{i}$

When $k = 2$: $z_2 = \sqrt{3}\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right) = -\frac{\sqrt{3}}{2} + \frac{3}{2}\text{i}$

When $k = 3$: $z_3 = \sqrt{3}(\cos\pi + \mathrm{i}\sin\pi) = -\sqrt{3}$

When $k = 4$: $z_4 = \sqrt{3}\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2} - \frac{3}{2}\mathrm{i}$

When $k = 5$: $z_5 = \sqrt{3}\left(\cos\frac{5\pi}{3} + \mathrm{i}\sin\frac{5\pi}{3}\right) = \frac{\sqrt{3}}{2} - \frac{3}{2}\mathrm{i}$

$\therefore z = \sqrt{3}, z = \frac{\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}, z = -\frac{\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}, z = -\sqrt{3}, z = -\frac{\sqrt{3}}{2} - \frac{3}{2}\mathrm{i}$ and $z = \frac{\sqrt{3}}{2} - \frac{3}{2}\mathrm{i}$

26) b)

$6 \times \frac{1}{2} \times \left(\sqrt{3}\right)^2 \times \sin\left(\frac{2\pi}{6}\right) = \frac{9\sqrt{3}}{2}$ units^2

26) c)

$\begin{pmatrix}\sqrt{3} & -1\\ 1 & \sqrt{3}\end{pmatrix}\begin{pmatrix}\sqrt{3}\\ 0\end{pmatrix} = \begin{pmatrix}3\\ \sqrt{3}\end{pmatrix}$

So $w_0 = 3 + \sqrt{3}\mathrm{i}$

$|w_0| = 2\sqrt{3}$ and $\arg w_0 = \frac{\pi}{6}$

$w_0^6 = \left(2\sqrt{3}\right)^6 \mathrm{e}^{\frac{6\pi}{6}\mathrm{i}} = -1728$

$w^6 + k = 0$

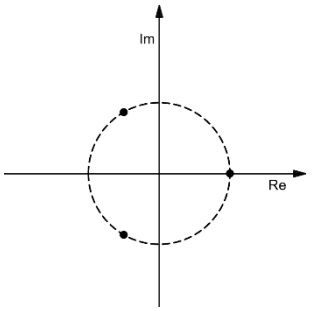
$\therefore k = 1728$

26) d)

$\det \mathbf{M} = \sqrt{3} \times \sqrt{3} + 1 \times 1 = 4$

So area after transformation is $4 \times \frac{9\sqrt{3}}{2} = 18\sqrt{3}$ units^2

27) a)



27) b)

The three cube roots of unity are equally spaced points on the unit circle forming a regular triangle centred at the origin. The centroid of a regular triangle is at the origin, so the vector sum of the three position vectors is zero.

27) c)

$z^3 = 1 + \mathrm{i}$

Using modulus-argument form: $1 + \mathrm{i} = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$

$r^3 = \sqrt{2} \Rightarrow r = \left(\sqrt{2}\right)^{\frac{1}{3}} = 2^{\frac{1}{6}}$

Since $3\theta = \frac{\pi}{4} + 2k\pi$ for integer k: $\theta = \frac{\pi}{12} + \frac{2k\pi}{3}$

When $k = 0$: $z_0 = 2^{\frac{1}{6}}\left(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\right)$

When $k = 1$: $z_1 = 2^{\frac{1}{6}}\left(\cos\frac{9\pi}{12} + \mathrm{i}\sin\frac{9\pi}{12}\right)$

When $k = 2$: $z_2 = 2^{\frac{1}{6}}\left(\cos\frac{17\pi}{12} + \mathrm{i}\sin\frac{17\pi}{12}\right) = 2^{\frac{1}{6}}\left(\cos\left(-\frac{7\pi}{12}\right) + \mathrm{i}\sin\left(-\frac{7\pi}{12}\right)\right)$

$\therefore z = 2^{\frac{1}{6}}\left(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\right)$, $z = 2^{\frac{1}{6}}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

and $z = 2^{\frac{1}{6}}\left(\cos\left(-\frac{7\pi}{12}\right) + \mathrm{i}\sin\left(-\frac{7\pi}{12}\right)\right)$

27) d)

Consider the sum of roots which is zero, so the imaginary part is also zero:

$2^{\frac{1}{6}}\sin\frac{\pi}{12} + 2^{\frac{1}{6}}\sin\frac{9\pi}{12} + 2^{\frac{1}{6}}\sin\left(-\frac{7\pi}{12}\right) = 0$

$\sin\frac{\pi}{12} + \sin\frac{9\pi}{12} + \sin\left(-\frac{7\pi}{12}\right) = 0$

$\sin 15° + \sin 135° + \sin(-105°) = 0$

But since sine is an odd function:

$\sin 15° + \sin 135° - \sin 105° = 0$

$\therefore \sin 15° + \sin 135° = \sin 105°$ as req.

28) a)

$z^4 = 2\sqrt{3} + 2\mathrm{i}$

Using modulus-argument form: $2\sqrt{3} + 2\mathrm{i} = 4\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$

$r^4 = 4 \Rightarrow r = 4^{\frac{1}{4}} = \sqrt{2}$

Since $4\theta = \frac{\pi}{6} + 2k\pi$ for integer k: $\theta = \frac{\pi}{24} + \frac{2k\pi}{4}$

When $k = 0$: $z_0 = \sqrt{2}\left(\cos\frac{\pi}{24} + \mathrm{i}\sin\frac{\pi}{24}\right)$

When $k = 1$: $z_1 = \sqrt{2}\left(\cos\frac{13\pi}{24} + \mathrm{i}\sin\frac{13\pi}{24}\right)$

When $k = 2$: $z_2 = \sqrt{2}\left(\cos\frac{25\pi}{24} + \mathrm{i}\sin\frac{25\pi}{24}\right) = \sqrt{2}\left(\cos\left(-\frac{23\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{23\pi}{24}\right)\right)$

When $k = 3$: $z_3 = \sqrt{2}\left(\cos\frac{37\pi}{24} + \mathrm{i}\sin\frac{37\pi}{24}\right) = \sqrt{2}\left(\cos\left(-\frac{11\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{11\pi}{24}\right)\right)$

$\therefore z = \sqrt{2}\left(\cos\frac{\pi}{24} + \mathrm{i}\sin\frac{\pi}{24}\right)$, $z = \sqrt{2}\left(\cos\frac{13\pi}{24} + \mathrm{i}\sin\frac{13\pi}{24}\right)$,

$z = \sqrt{2}\left(\cos\left(-\frac{23\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{23\pi}{24}\right)\right)$ and $z = \sqrt{2}\left(\cos\left(-\frac{11\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{11\pi}{24}\right)\right)$

28) b)

Consider the sum of roots which is zero, so the real part is also zero:

$\sqrt{2}\cos\frac{\pi}{24} + \sqrt{2}\cos\frac{13\pi}{24} + \sqrt{2}\cos\left(-\frac{23\pi}{24}\right) + \sqrt{2}\cos\left(-\frac{11\pi}{24}\right) = 0$

$\cos\frac{\pi}{24} + \cos\frac{13\pi}{24} + \cos\left(-\frac{23\pi}{24}\right) + \cos\left(-\frac{11\pi}{24}\right) = 0$

$\cos 7.5° + \cos 97.5° + \cos(-172.5°) + \cos(-82.5°) = 0$

But since cosine is an even function:

$\cos 7.5° + \cos 97.5° + \cos 172.5° + \cos 82.5° = 0$

$\therefore \cos 7.5° + \cos 82.5° + \cos 97.5° + \cos 172.5° = 0$ as req.

29) a)

$z^5 = 3^5$ so $z^5 = 243$ or $z^5 - 243 = 0$

29) b)

$z^5 = 243$

Using modulus-argument form: $243 = 243(\cos 0 + \mathrm{i}\sin 0)$

$z^5 = r^5(\cos\theta + \mathrm{i}\sin\theta)^5 = r^5(\cos 5\theta + \mathrm{i}\sin 5\theta)$

$r = 3$

Since $5\theta = 0 + 2k\pi$ for integer k: $\theta = 0 + \frac{2k\pi}{5}$

When $k = 0$: $z_0 = 3(\cos 0 + \mathrm{i}\sin 0) = 3$

When $k = 1$: $z_1 = 3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)$

When $k = 2$: $z_2 = 3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right)$

When $k = 3$: $z_3 = 3\left(\cos\frac{6\pi}{5} + \mathrm{i}\sin\frac{6\pi}{5}\right) = 3\left(\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right)$

When $k = 4$: $z_4 = 3\left(\cos\frac{8\pi}{5} + \mathrm{i}\sin\frac{8\pi}{3}\right) = 3\left(\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right)$

$\therefore z = 3$, $z = 3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)$, $z = 3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right)$,

$z = 3\left(\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right)$ and $z = 3\left(\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right)$

29) c)

$z_B = 3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)$ and $z_E = 3\left(\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right)$

Summing z_B and z_E:

$3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right) + 3\left(\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right)$

$= 3\left[\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5} + \cos\left(-\frac{2\pi}{5}\right) - \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right]$

$= 3\left[\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5} + \cos\frac{2\pi}{5} - \mathrm{i}\sin\frac{2\pi}{5}\right]$

$= 3 \times 2\cos\frac{2\pi}{5}$

Multiplying z_B and z_E:

$3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right) \times 3\left(\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right)$

$= 9\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)\left(\cos\frac{2\pi}{5} - \mathrm{i}\sin\frac{2\pi}{5}\right)$

$= 9\left(\cos^2\frac{2\pi}{5} + \sin^2\frac{2\pi}{5}\right)$

$= 9$

So that $z^2 - 6z\cos\frac{2\pi}{5} + 9 = 0$ as req.

29) d)

$z_C = 3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right)$ and $z_D = 3\left(\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right)$

Summing z_C and z_D:

$3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right) + 3\left(\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right)$

$= 3\left[\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5} + \cos\left(-\frac{4\pi}{5}\right) - \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right]$

$= 3\left[\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5} + \cos\frac{4\pi}{5} - \mathrm{i}\sin\frac{4\pi}{5}\right]$

$= 3 \times 2\cos\frac{4\pi}{5}$

Multiplying z_c and z_D:

$3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right) \times 3\left(\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right)$

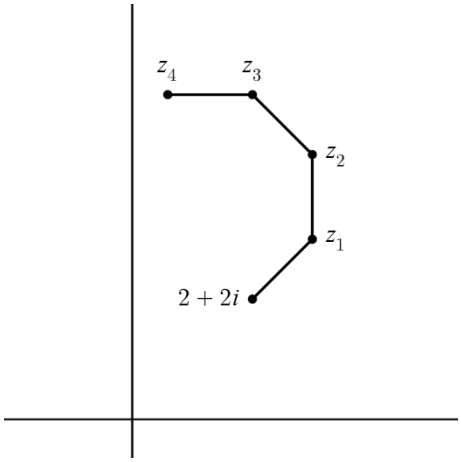
$= 9\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right)\left(\cos\frac{4\pi}{5} - \mathrm{i}\sin\frac{4\pi}{5}\right)$

$= 9\left(\cos^2\frac{4\pi}{5} + \sin^2\frac{4\pi}{5}\right)$

$= 9$

So that $z^2 - 6z\cos\frac{4\pi}{5} + 9 = 0$

30)



Set the problem to start at the origin and consider the vectors:

$v_1 = \sqrt{2}\mathrm{e}^{\frac{\pi}{4}\mathrm{i}} = 1 + \mathrm{i}$

$v_2 = \sqrt{2}\mathrm{e}^{\frac{\pi}{2}\mathrm{i}} = \sqrt{2}\mathrm{i}$

$v_3 = \sqrt{2}\mathrm{e}^{\frac{3\pi}{4}\mathrm{i}} = -1 + \mathrm{i}$

$v_4 = \sqrt{2}\mathrm{e}^{\pi\mathrm{i}} = -\sqrt{2}$

The sum: $1 + \mathrm{i} + \sqrt{2}\mathrm{i} - 1 + \mathrm{i} - \sqrt{2} = -\sqrt{2} + \left(2 + \sqrt{2}\right)\mathrm{i}$

Now consider the displacement from $2 + 2\mathrm{i}$:

$2 + 2\mathrm{i} - \sqrt{2} + \left(2 + \sqrt{2}\right)\mathrm{i} = 2 - \sqrt{2} + \left(4 + \sqrt{2}\right)\mathrm{i}$

31) a)

Solve $w^2 = 8 - 8\sqrt{3}\mathrm{i}$ where $w = z - 1$

Using modulus-argument form: $8 - 8\sqrt{3}\mathrm{i} = 16\left(\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)$

$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta)$

$r^2 = 16 \Rightarrow r = 4$

Since $2\theta = -\frac{\pi}{3} + 2k\pi$ for integer k: $\theta = -\frac{\pi}{6} + \frac{2k\pi}{2}$

When $k = 0$: $z = 4\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right) = 2\sqrt{3} - 2\mathrm{i}$

When $k = 1$: $z = 4\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right) = -2\sqrt{3} + 2\mathrm{i}$

So $w = 2\sqrt{3} - 2\mathrm{i}$ and $w = -2\sqrt{3} + 2\mathrm{i}$ and so $z = w + 1$

$\therefore z = 1 + 2\sqrt{3} - 2\mathrm{i}$ and $z = 1 - 2\sqrt{3} + 2\mathrm{i}$

31) b)

$w = z - 1$ so $|w|^2 = |z - 1|^2$ which gives

$\left|2\sqrt{3} - 2\mathrm{i}\right|^2 = 16$ and $\left|-2\sqrt{3} + 2\mathrm{i}\right|^2 = 16$ units.

For both roots it is 16 units.

31) c)

32 units.

31) d)

$|z_1|^2 + |z_2|^2$

$= \left|2\sqrt{3} - 2\mathrm{i}\right|^2 + \left|-2\sqrt{3} + 2\mathrm{i}\right|^2$

$= 17 + 4\sqrt{3} + 17 - 4\sqrt{3} = 34$ units.

Alternatively:

$32 + 1 + 1 = 34$ units.

32) a)

Solve $w^2 = -40 + 40\sqrt{3}\text{i}$ where $w = z - 3$

Using modulus-argument form: $40 - 40\sqrt{3}\text{i} = 80\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right)$

$z^2 = r^2(\cos\theta + \text{i}\sin\theta)^2 = r^2(\cos 2\theta + \text{i}\sin 2\theta)$

$r^2 = 80 \Rightarrow r = \sqrt{80}$

Since $2\theta = \frac{2\pi}{3} + 2k\pi$ for integer k: $\theta = \frac{2\pi}{6} + \frac{2k\pi}{2}$

When $k = 0$: $z_0 = \sqrt{80}\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right) = -2\sqrt{5} + 2\sqrt{15}\text{i}$

When $k = 1$: $z_1 = \sqrt{80}\left(\cos\frac{4\pi}{3} + \text{i}\sin\frac{4\pi}{3}\right) = -2\sqrt{5} - 2\sqrt{15}\text{i}$

So $w = -2\sqrt{5} + 2\sqrt{15}\text{i}$ and $w = -2\sqrt{5} - 2\sqrt{15}\text{i}$ and so $z = w + 2$

$\therefore z = 3 + 2\sqrt{5} - 2\sqrt{15}\text{i}$ and $z = 3 + 2\sqrt{5} + 2\sqrt{15}\text{i}$

32) b)

$r = \sqrt{80}$ so $|z - 3|^2 = 80$ units.

32) c)

160 units.

32) d)

$|z_1|^2 + |z_2|^2$

$= \left|3 + 2\sqrt{5} + 2\sqrt{15}\text{i}\right|^2 + \left|3 - 2\sqrt{5} - 2\sqrt{15}\text{i}\right|^2$

$= 89 + 12\sqrt{5} + 89 - 12\sqrt{5} = 178$ units.

Alternatively:

$160 + 9 + 9 = 178$ units.

33)

$|z - 2|^2 = (z - 2)(z - 2)^* = (z - 2)(z^* - 2) = zz^* - 2z - 2z^* + 4$

$= |z|^2 - 2(z + z^*) + 4$

If ω is any complex n^{th} roots of unity then $|\omega|^2 = 1$

So that $|\omega - 2|^2 = 1 - 2(\omega + \omega^*) + 4 = 5 - 2(\omega + \omega^*)$ as req.

34) a)

$|z-3|^2 = (z-3)(z-3)^* = (z-3)(z^*-3) = zz^* - 3z - 3z^* + 9$

$= |z|^2 - 3(z+z^*) + 9$

If ω is any complex n^{th} roots of unity then $|\omega|^2 = 1$

So that $|\omega - 3|^2 = 1 - 3(\omega + \omega^*) + 9 = 10 - 3(\omega + \omega^*)$ as req.

34) b)

$$\sum_{k=0}^{4} \left|3-\omega^k\right|^2 = \sum_{k=0}^{4} \left(10 - 3(\omega^k + (\omega^k)^*)\right) = 50 - 3\sum_{k=0}^{4}(\omega^k + (\omega^k)^*)$$

Since $\sum_{k=0}^{4} \omega^k = 0$ and $\sum_{k=0}^{4}(\omega^k)^* = 0$

$\therefore (PA)^2 + (PB)^2 + (PC)^2 + (PD)^2 + (PE)^2 = 50$

35) a)

Since $3\theta = 0 + 2k\pi$ for integer k: $\theta = 0 + \frac{2k\pi}{3}$ which gives:

$ae^{0i} = a$, $ae^{\frac{2\pi}{3}i}$ and $ae^{-\frac{2\pi}{3}i}$

35) b)

$a(1+\omega+\omega^2) = a \times 0 = 0$

35) c)

$M_{AB} = \frac{A+B}{2} = \frac{a(1+\omega)}{2}$, $M_{BC} = \frac{B+C}{2} = \frac{a(\omega+\omega^2)}{2}$, $M_{CA} = \frac{C+A}{2} = \frac{a(\omega^2+1)}{2}$

Since $1 + \omega + \omega^2 = 0$

$M_{AB} = -\frac{1}{2}a\omega^2$, $M_{BC} = -\frac{1}{2}a$, $M_{CA} = \frac{C+A}{2} = -\frac{1}{2}a\omega$

Which are the three cube-roots of unity scaled by $-\frac{1}{2}a$

$(z-p)(z-p\omega)(z-p\omega^2)$

$= (z^2 - p(1+\omega)z + p^2\omega)(z - p\omega^2)$

$= z^3 - p(1+\omega)z^2 + p^2\omega z - p\omega^2 z^2 + p^2(\omega^2+\omega^3)z - p^3\omega^3$

Since $\omega^3 = 1$:

$= z^3 - p(1+\omega)z^2 + p^2\omega z - p\omega^2 z^2 + p^2(\omega^2+1)z - p^3$

$= z^3 - p(1+\omega+\omega^2)z^2 + p^2(\omega^2+\omega+1)z - p^3$

$= z^3 - p^3$

In this case, $p = -\frac{1}{2}a$

And so $z^3 - p^3 = z^3 - \left(-\frac{1}{2}a\right)^3 = z^3 + \frac{a^3}{8}$

$\therefore 8z^3 + a^3 = 0$

36) a)

Since $5\theta = 0 + 2k\pi$ for integer k: $\theta = 0 + \frac{2k\pi}{5}$ which gives:

ae^{0i}, $ae^{\frac{2\pi}{5}i}$, $ae^{\frac{4\pi}{5}i}$, $ae^{\frac{6\pi}{5}i}$, $ae^{\frac{8\pi}{5}i}$

ae^{0i} a, $ae^{\frac{2\pi}{5}i}$, $ae^{\frac{4\pi}{5}i}$, $ae^{-\frac{4\pi}{5}i}$ and $ae^{-\frac{2\pi}{5}i}$

36) b)

$a(1 + \omega + \omega^2 + \omega^3 + \omega^4) = a \times 0 = 0$

36) c)

Since $e^{i\theta} = \cos\theta + i\sin\theta$ then $\cos x = \frac{1}{2}\left(e^{ix} + e^{-ix}\right)$

So $\cos\left(\frac{\alpha-\beta}{2}\right) = \frac{1}{2}\left(e^{\left(\frac{\alpha-\beta}{2}\right)i} + e^{-\left(\frac{\alpha-\beta}{2}\right)i}\right)$

$\Rightarrow 2\cos\left(\frac{\alpha-\beta}{2}\right) = e^{\left(\frac{\alpha-\beta}{2}\right)i} + e^{-\left(\frac{\alpha-\beta}{2}\right)i}$

$\therefore 2\cos\left(\frac{\alpha-\beta}{2}\right)e^{\frac{\alpha+\beta}{2}i} = e^{\frac{\alpha+\beta}{2}i}\left(e^{\left(\frac{\alpha-\beta}{2}\right)i} + e^{-\left(\frac{\alpha-\beta}{2}\right)i}\right) = e^{\frac{\alpha+\beta}{2}i}e^{\left(\frac{\alpha-\beta}{2}\right)i} + e^{\frac{\alpha+\beta}{2}i}e^{-\left(\frac{\alpha-\beta}{2}\right)i}$

$= e^{\frac{2\alpha}{2}i} + e^{\frac{2\beta}{2}i} = e^{\alpha i} + e^{\beta i}$ as req.

36) d)

A general midpoint is given by $\frac{a\omega^k + a\omega^{k+1}}{2}$

$M_{AB} = \frac{A+B}{2}$, $M_{BC} = \frac{B+C}{2}$, etc

From part c)

$M_{AB} = \frac{ae^{\alpha i} + ae^{\beta i}}{2} = a\cos\left(\frac{\alpha-\beta}{2}\right)e^{\frac{\alpha+\beta}{2}i}$

When $\alpha = 0$, $\beta = \frac{2\pi}{5}$ then $M_{AB} = a\cos\left(\frac{\pi}{5}\right)e^{\frac{\pi}{5}i}$

Similarly for the other midpoints.

The original roots satisfy $z^5 - a^5 = 0$

There is a scale factor of $\cos\left(\frac{\pi}{5}\right)$ so for any midpoint, w then $w = z\cos\frac{\pi}{5}$

So $z^5 - a^5 = 0$ gives $\left(\frac{w}{\cos\frac{\pi}{5}}\right)^5 - a^5 = 0$

$\therefore w^5 - \cos^5\left(\frac{\pi}{5}\right)a^5 = 0$ as req.

36) e)

$w^6 - \cos^6\left(\frac{\pi}{6}\right)a^6 = 0$

$\Rightarrow w^6 - \frac{27}{64}a^6 = 0$

$\therefore 64w^6 - 27a^6 = 0$